

Dylan Spektor

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EDUCATION & HONORS

University of Colorado Boulder

Expected: December 2027

B.S. Applied Mathematics and B.A. Computer Science

- **GPA:** 4.0
- **Organizations:** CU Quants, Engineering Honors Program, Tau Beta Pi, Quantum Scholars, HackCU.
- **Relevant Coursework:** Mathematical Finance, Advanced Data Science, Software Development Methods, Algorithms, Probability and Statistics, Numerical Methods and Scientific Computing, Applied Real Analysis.

Colorado Early Colleges Douglas County North

May 2025

AS. Computer Science (Dual Enrollment at Arapahoe Community College)

- **GPA:** 4.85 (weighted)
- **Honors & Awards:** 1580 SAT, National Merit Finalist, top 5% rank, CyberPatriot Platinum Tier Competitor.

RESEARCH & QUANTITATIVE EXPERIENCE

CU Quants Fund

Oct 2025 – Present

Quantitative Developer

- Build and optimize high-performance trading infrastructure, developing robust systems for strategy execution, market data processing, and real-time risk monitoring.
- Perform data analysis on historical orderbook data, modeling spread of stablecoin assets to identify market making opportunities. Target liquidity provision for large institutional flow outside immediate top-of-book.
- Explore cross-correlation analysis to identify leader-follower relationships in cryptocurrency markets, and apply convex optimization to solve for optimal capital allocation across pairs trading strategies.

Discovery Learning Apprentice, Automated Reasoning Research

Aug 2026 – Apr 2027

Advised by Prof. Evan Chang, CU Boulder

- Selected for a yearlong, paid research apprenticeship developing algorithmic methods for automated reasoning and formal verification of software systems, leveraging functional programming and rigorous mathematical reasoning.

PROJECTS

Pairs Trading Strategy Research

April 2026

- Researched and backtested a statistical arbitrage strategy on 497 S&P 500 equities (10 years of data), using PCA and weighted PageRank to filter 124,750 candidate pairs down to 300 for Engle-Granger cointegration testing. Built a regime-adaptive signal combining GMM volatility clustering with MinHash historical pattern-matching, then trained a Random Forest classifier that lifted signal precision to 0.97/0.84/0.70 F1 (exit/long/short) out-of-sample, generating ~6x the cumulative return of the raw regime+pattern signal in backtesting.

Skills: Python (pandas, numpy, scikit-learn, statsmodels, datasketch), machine learning, time series analysis

SMT-Based Cyclic Arbitrage Detection

July 2026

- Built a Z3 SMT-based cross-currency arbitrage detector as a more expressive alternative to Bellman-Ford, validating exact agreement on a frictionless baseline before extending to fixed fees, nonlinear slippage, and integer lot sizes. Uncovered a capital-dependent profitability window (breakeven at 0.506, peak profitability at 1.2–1.6x starting capital) invisible to Bellman-Ford, confirmed via brute-force enumeration, while benchmarking Z3 as ~500x slower than Bellman-Ford in the frictionless case.

Skills: Python, Z3/SMT solvers, graph algorithms, constraint solving, algorithm design, quantitative modeling

Zeroes of the Riemann Zeta Function and Spectral Statistics of Random Matrices

April 2026

- Co-authored a complex analysis report connecting Riemann zeta zeros to Gaussian/Circular Unitary Ensemble spectral statistics from random matrix theory, exploring the Hilbert–Pólya conjecture. Derived moment-based connections between random matrix and zeta statistics and outlined a contour-integration approach to a key zero-spacing conjecture.

Skills: Mathematical research, statistics, random matrix theory, technical writing & presentation

External Quantum Efficiency Degradation in Perovskite Cells

December 2025

- An experimental physics research collaboration with the National Laboratory of the Rockies investigating how C60 transport-layer deposition cycles affect perovskite solar-cell EQE degradation; analyzed weekly 350–400 nm EQE measurements across 0, 5, and 10 deposition cycles, using 10,000× bootstrap resampling within measurement uncertainty and quadratic/linear regression to estimate degradation-rate distributions, with ANOVA showing statistically significant differences in both linear and quadratic degradation coefficients across deposition conditions.

Skills: Python (pandas, numpy, scipy, scikit-learn), statistical modeling, hypothesis testing, data analysis